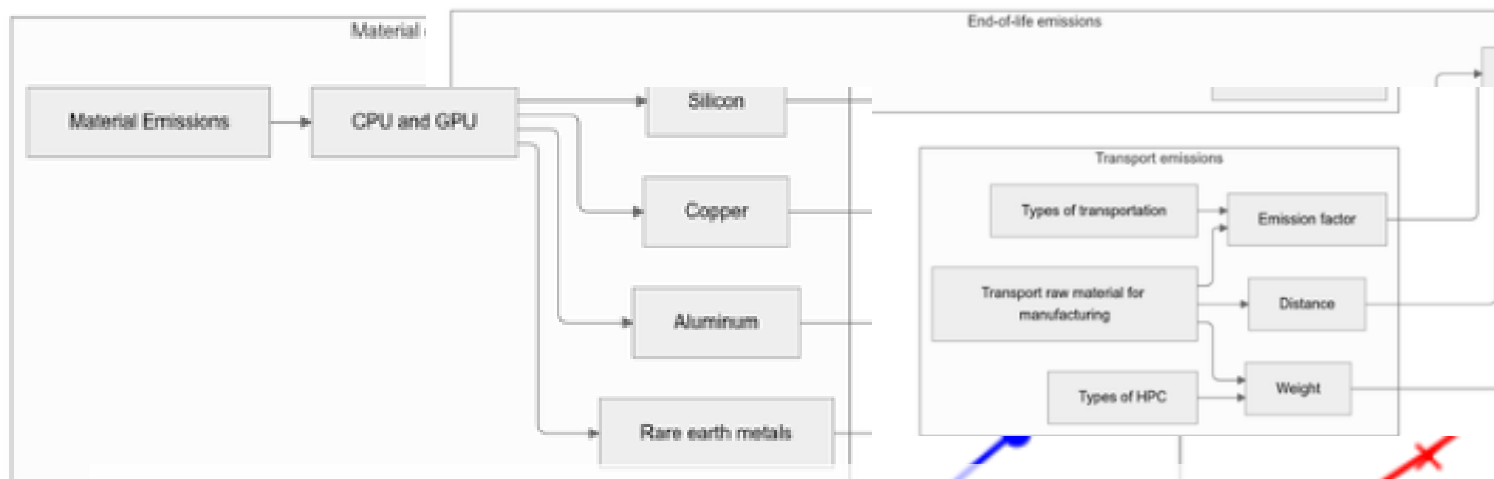
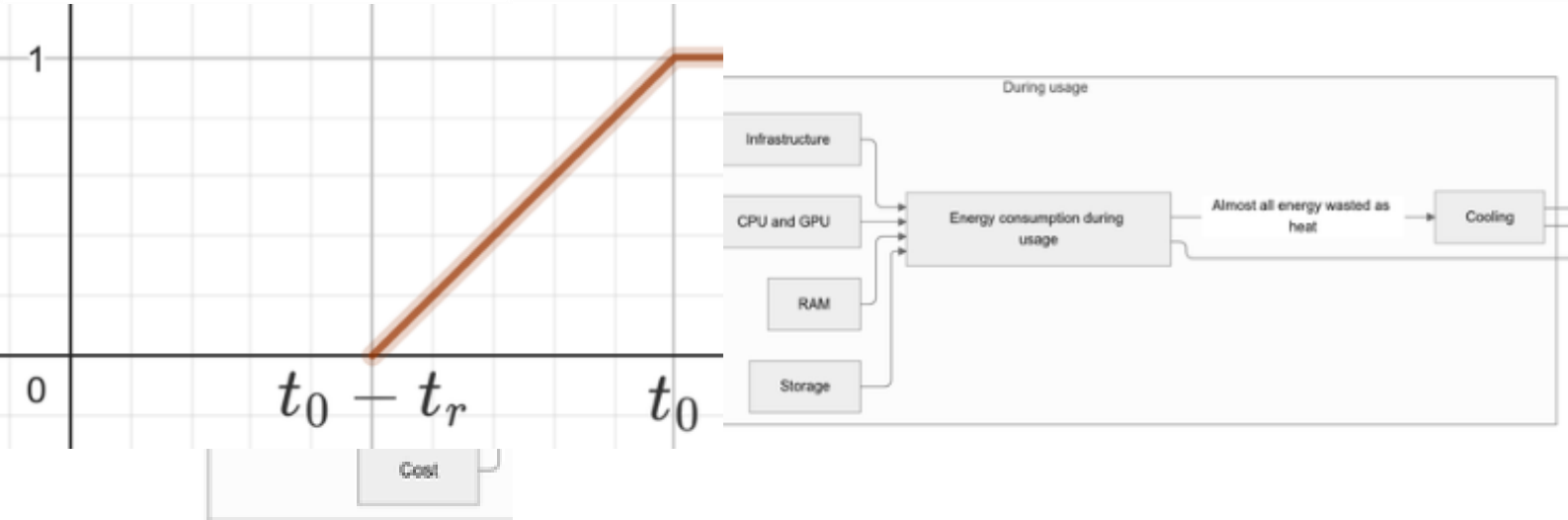


High-performance computing (HPC) systems are integral to advancing scientific research, engineering, and industry, but their environmental impact—particularly regarding carbon emissions and water usage—is a growing concern. Our research examines the projected carbon dioxide emissions and water consumption of HPC systems globally, with a focus on projections for 2030. We use a variety of models, including linear and polynomial regressions, moving average methods, and hardware-specific scaling factors, to predict future emissions based on historical data on power consumption, electricity generation sources, and carbon emission factors.

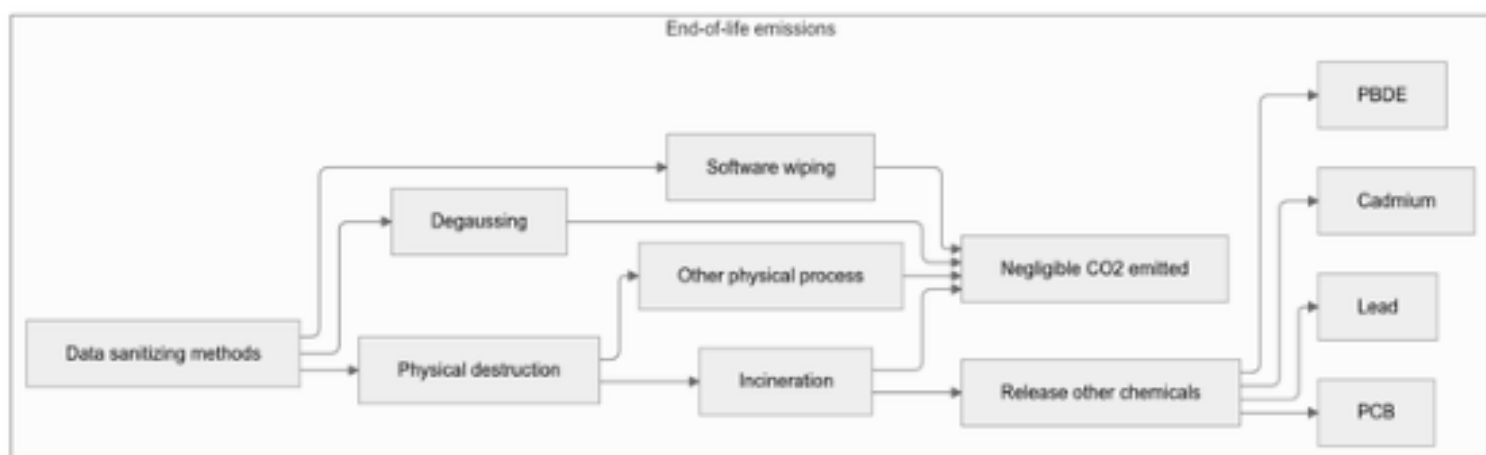
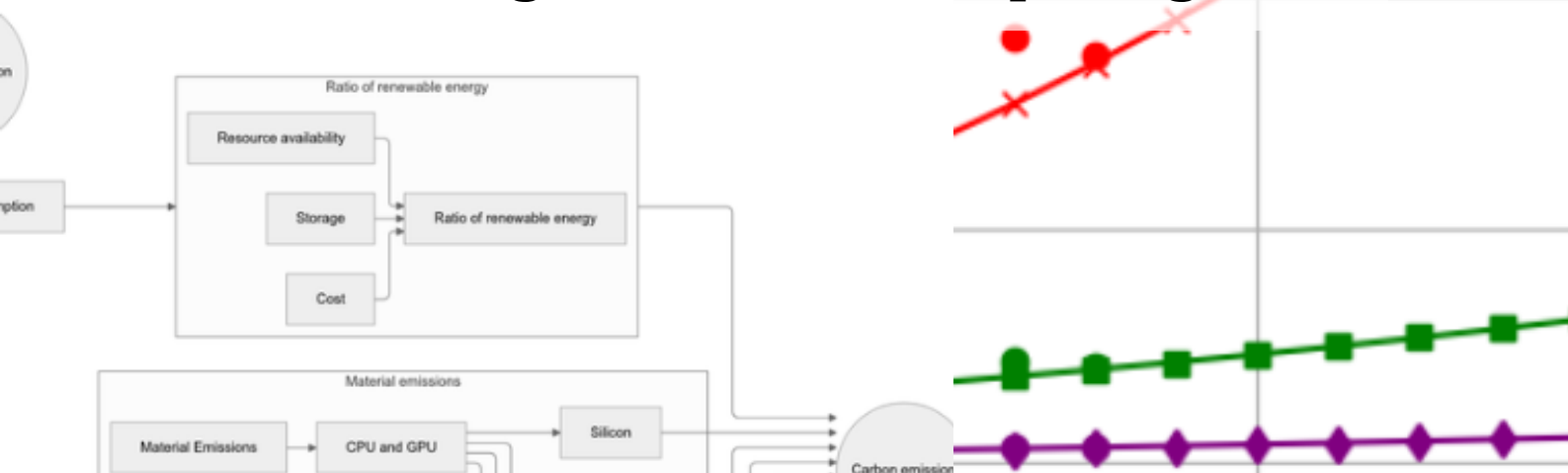
Our approach divides the life cycle of HPC systems into four major stages: material manufacturing, transportation, usage, and end-of-life. This detailed segmentation helps to pinpoint where environmental impacts arise, making it easier for us to target interventions at each stage. By incorporating this granular analysis, our models offer more precise recommendations for policies and technical improvements, ensuring a comprehensive approach to sustainability.

A key innovation in our model is its treatment of the usage phase, where we integrate moving averages for power consumption trends across CPU, GPU, and RAM hardware. This takes into account the replacement cycle of chip generations and the gradual transition between hardware generations in large-scale HPC deployments. As a result, our model offers a more accurate representation of energy demand over time, distinguishing it from others that may overlook these transitional periods.

This research advocates for strategies like optimizing energy efficiency during operation and transitioning to renewable energy sources, as these will have the most significant long-term impact on reducing emissions. Finally, we predict the trends of carbon emissions of HPC using polynomial regression lines and calculate the possible amount, which is approximately 385.05643 million metric tons.



Examining the Environmental Impact of High-Powered Computing



Contents

Introduction	2	Software Improvements	18
Background	2	Waste Heat Reuse and Water Reuse . .	18
Problem restatement	2	Geographical Flexibility	18
Assumptions and Justifications	2	Conclusions	19
Variables	3	References	19
2.1 Material stage	3	Letter to the Advisory Board	21
2.2 Transportation stage	3	Appendices	23
2.3 Computation stage	4	Codes	23
2.4 End-of-life stage	4	Report on the use of AI	24
2.5 Refined model variables	4		
Carbon Emission Models	4		
Material Emission Model	5		
Transport Emission Model	6		
Usage Emission Model	8		
Energy consumption of cooling . .	8		
End-of-life Emission Model	8		
Apply the Model	9		
During Material Stage	9		
During Transportation Stage	10		
During Usage Stage	11		
Finding $\vec{k}_{\text{carbon}}$	11		
Predicting $\vec{r}_{\text{energy}}$ in the future . . .	12		
Modelling $E_{\text{computation}}$	12		
Predicting C_{usage}	13		
During End-of-Life Stage	14		
Result of the Model	14		
Refine Previous Model	14		
Water Usage	14		
CPU, GPU, and RAM Power Consumption Trends	14		
Apply Refined Model	15		
Future trends	15		
How model changes if HPC grows .	15		
How model changes if energy demanded in other sectors grows	15		
How model changes if energy sources change	15		
Predicting $\text{Water}_{\text{total}}$ in future	15		
Impact of replacing old hardware faster .	16		
Strengths and Weaknesses	16		
Strengths	16		
Weaknesses	17		
Recommendations to reduce environmental impact	17		
Hardware Optimization	18		

Introduction

Background

High-Powered Computing (HPC) plays a pivotal role in advancing scientific research, technological innovation, and industry applications. HPC systems are used in fields ranging from climate modeling and data collection to artificial intelligence and engineering simulations. These systems often consist of thousands or even millions of processors, making them exceptionally powerful but also energy-intensive.

The U.S. International Trade Commission stated that in 2010, there was reportedly 1.2 trillion gigabytes of new data created globally, with this number projected to increase to 175 trillion gigabytes by 2025. (Andy_Patrizio, 2018) As the demand for HPC capabilities continues to grow, the environmental impact—particularly in terms of energy consumption—has become an increasing concern.

Globally, the energy consumption of HPC systems is substantial. The total energy usage of the world's top supercomputers has been estimated in the tens of terawatt-hours (TWh) annually. For example, as of 2023, the top 500 supercomputers in the world collectively consume an estimated 50 to 100 TWh of electricity per year. (*Green HPC: An Analysis of the Domain Based on Top500*, n.d.) While this represents a small percentage of total global energy use, the efficiency of these systems varies significantly depending on factors such as hardware specifications, cooling mechanisms, and utilization rates. (McInnes et al., 2021)

The energy consumption of HPC systems is typically categorized into two components: full capacity usage and average utilization rates. Full capacity refers to the maximum energy demand when the systems are running at peak performance, while average utilization rates represent the typical operational energy usage during periods of active computation. Supercomputers are often designed for specific, high-intensity tasks that may not run at full capacity at all times. For example, systems in use for research or simulation may only operate at peak load during certain phases of a project, resulting in periods of lower average utilization. Nevertheless, the sheer scale of HPC infrastructures, which often remain powered on and ready for rapid deployment, ensures that their

energy consumption remains high.

Despite the advancements in energy-efficient hardware and cooling technologies, the overall carbon footprint of HPC remains significant, especially when many of these systems are powered by conventional energy sources such as coal and natural gas. Furthermore, the increasing integration of artificial intelligence (AI) and machine learning workloads, which require immense computational power, adds to the strain on energy resources. This growing energy demand raises concerns about the long-term sustainability of HPC systems and their environmental consequences, prompting a need for more sustainable practices and innovations in energy-efficient computing solutions.

Problem restatement

As the field of HPC continues to expand, understanding its environmental footprint—specifically its e-waste, water consumption, and overall energy usage—is critical for developing strategies to mitigate its impact on global energy consumption and climate change (Smith et al., 2021). This work aims to explore the environmental implications of HPC, focusing on models of energy usage, efficiency improvements, and a comparison of strengths and weaknesses between various approaches (Jones & Patel, 2020). Furthermore, the role of sustainable practices in reducing the ecological footprint of high-powered computing is increasingly recognized, with emerging initiatives to optimize computational efficiency and minimize waste (Williams, 2022).

Assumptions and Justifications

We made the following assumptions to simplify our model to a state that could be completed in the given time-frame:

1.3.1 Assumption 1: CPUs and GPUs in HPC only consist of silicon, copper, aluminum, and rare earth metals.

In the model of material emission, we consider silicon, copper, aluminum, and rare earth metals as basic materials to calculate the carbon emissions produced. This assumption can be justified based on their roles in construction and functionality. First, silicon is the foundational material for semiconductors, used to create transistors, diodes, and

integrated circuits. It contributes to the largest proportion of carbon emissions. Copper is primarily used for electrical interconnections within CPUs and GPUs, amounting to around 5-10% of the total mass of a chip. Aluminum accounts for around 1-5% of the total weight of CPUs and GPUs. Rare earth elements are utilized in specific components such as magnets in GPUs and certain capacitors. (Euromines, 2020)

While other materials (like ceramics, or specialized alloys) may be used in packaging or minor components, their contributions to the overall emissions are negligible compared to silicon, copper, aluminum, and rare earth metals.

1.3.2 Assumption 2: Weights of all elements used to count for an average are the same.

Without this assumption, we need to spend a lot of time searching for data, computing weight of these elements. We assume this for the convenience of calculation.

1.3.3 Assumption 3: Carbon emissions from HPC hardware destruction are negligible compared to overall computing emissions.

Every server has a limited lifespan, and most servers can't function normally after being utilized for 6 to 10 years. After this period of use, the servers and hardware have to be processed properly. Inappropriate e-waste disposal contributes significantly to contamination, lead release, and overuse of land or space. From a security perspective, data thieves can filch information in discarded hardware if not properly sanitized. Accordingly, data wiping and suitable methods for recycling are crucial. Data sanitizing methods include software wiping, degaussing, and physical destruction.

Firstly, software wiping aims to overwrite all data by some special software, making it unrecoverable. This is obviously carbon-free, considering only the use of software (Verity Systems). Secondly, degaussing works by leaving magnetic domains, which are integral parts in data storing, randomly disorientated. Therefore, all magnetic-stored data could be erased completely. This disruption and elimination method hardly produce any carbon directly (Jones & Davis, 2018). It then finally comes to the method of physically destroying the hardware. Commonly used ways are shredding, hammering, corrosion, melting, and incineration.

From the above, incineration has a relatively more significant impact on the environment, releasing cadmium, lead, mercury, nickel, tin, PBDEs, and PCBs (George, R. 2024). However, this yields little carbon that could be omitted when compared to the major carbon producers in HPC. Other measures that should be adopted after wiping data are recycling or reusing hardware components.

To sum up, carbon emitted during the destruction of HPC hardware is insignificant and ignorable.

Variables

Before introducing the models that we developed, listing out and elucidating the variables used in the equation will be helpful. Our model for calculating carbon could be expressed by the sum of several small models. It contains the following major variables:

Generally, we have:

- C_{total} , the total CO₂ emitted in HPC annually.

2.1 Material stage

- C_{mat} , the total CO₂ emitted during the obtaining, synthesis and integration of all materials needed for building hardware. (in metric tons)
- $M_{\text{mat } i}$, the total mass of material i used in the construction of HPC hardware. (in metric tons)
- $ef_{\text{mat } i}$, the emission factor of material i , measuring the amount of CO₂ released by i per unit weight. (in metric tons per metric tons of i)

2.2 Transportation stage

- C_{trans} , the total CO₂ emitted during the transportation of materials. (in metric tons)
- D_{trans} , the total distance travelled by raw materials of HPC hardware. (in kilometres)
- $M_{\text{raw } i}$, the mass of raw material i used to build HPC hardware. (in metric tons)
- ef_{trans} , the emission factor of transportation, measuring the amount of CO₂ released per

unit weight per unit length. (in metric tons per kilometre per kilogram)

- $M_{\text{multi GPU}}$, mass of multi-GPU high-powered computer. (in metric tons)
- $M_{\text{single GPU}}$, mass of a single GPU HPC high-powered computer. (in metric tons)
- $n_{\text{GPU per node}}$, the number of GPUs on one node in average.
- n_{node} , the number of nodes in HPC.

2.3 Computation stage

- C_{usage} , the total CO_2 emitted during computation (in metric tons).
- $P_{\text{computation}}$, the average power consumed globally by HPC systems. (in kW)
- $E_{\text{computation}}$, the total energy consumed during computation from non-renewable sources. (in kWh when considering derivative relationship with $P_{\text{computation}}$, in TWh at last to ensure smaller and readable number)
- $\vec{r}_{\text{energy}}$, the global percentage of electricity generated by each source, in the form of a vector.
- $\vec{k}_{\text{carbon}}$, the amount of CO_2 emitted per watt-hour of electricity generated using each source mentioned in $\vec{r}_{\text{energy}}$, in the form of a vector. It is considered to have a constant value.
- $k_{\text{cooling energy}}$, energy used for cooling per unit of heat emitted by the HPC system, is a ratio and has no unit.

2.4 End-of-life stage

- C_{end} , the CO_2 emitted when destroying the hardware. (in metric tons)

2.5 Refined model variables

- $\text{Water}_{\text{usage}}$, the water consumption during usage. (in million metric tons)

Notice that metric tons is often abbreviated as *MT* or *tCO₂e* when describing specifically the amount of CO_2 emission and that million metric tons could be denoted by *MMT*. A few variables use kilogram or kg, so we will carefully calculate not only the number but also the unit.

Carbon Emission Models

The carbon emission of HPC can be assessed in different aspects, which can be expressed by using the following formula:

$$C_{\text{total}} = \sum C = C_{\text{mat}} + C_{\text{trans}} + C_{\text{usage}} + C_{\text{end}} \quad (1)$$

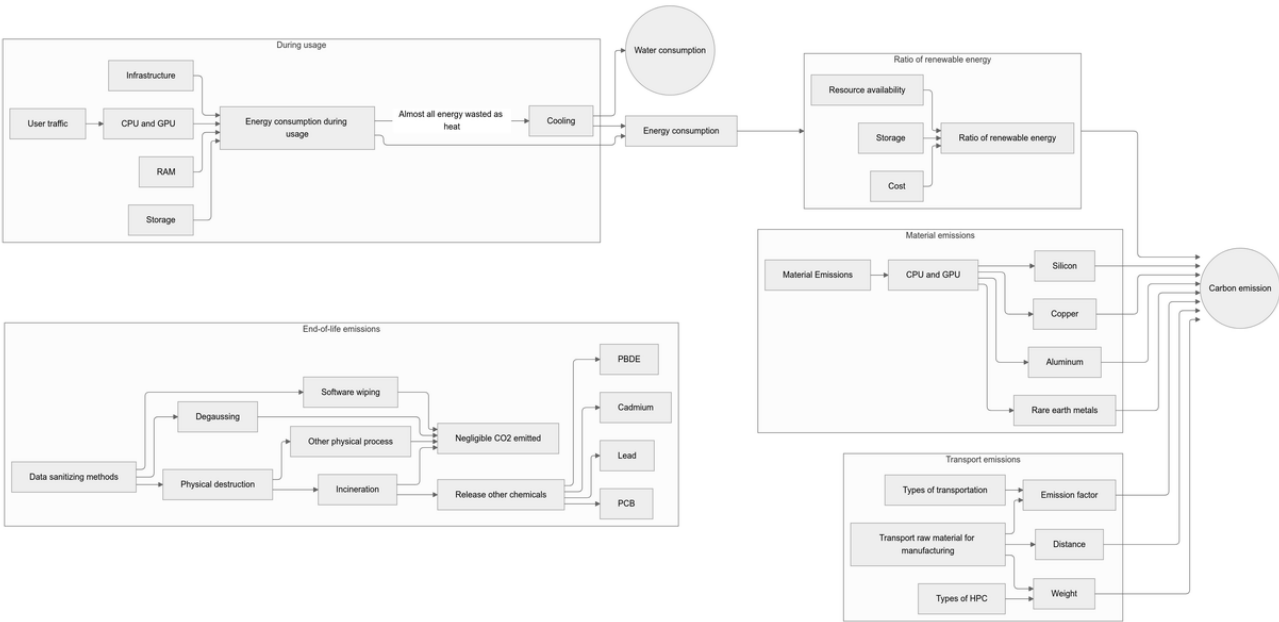
Where:

Material Emissions: Based on the types and amounts of materials used in the hardware (metals, rare earth elements, etc.), their extraction, modification, and manufacture.

Transport Emissions: Emissions generated from transporting raw materials to the manufacturing facility and the finished product to market.

Usage Emissions: Energy used while running the HPC system.

End-of-life Emissions: Emissions produced when the hardware reaches the end of its useful life (e.g., disposal, recycling, or landfill).



Graph 3.0.0.1

Material Emission Model

The **material carbon emissions** associated with HPC manufacturing can be calculated by the formula below:

$$C_{\text{mat}} = \sum_i (M_{\text{mat } i} \cdot ef_{\text{mat } i})$$

Where C_{mat} refers to the total carbon emissions during the material production process in metric tons, $M_{\text{mat } i}$ represents the mass of material i (certain material) used in the manufacturing process (in metric tons), $ef_{\text{mat } i}$ stands for the emission factor for material i (in MT of CO₂ per metric ton of material) (*Calculating Emissions / Minnesota Pollution Control Agency*, n.d.)

This formula enables the aggregation of carbon emissions from the various materials employed in the manufacturing process by multiplying the mass of each material by its corresponding emission factor. As the emission factors for each material have

been rigorously determined through scientific studies, we have directly referenced these established statistics. (*General Conformity Training Modules: Appendix a Sample Emissions Calculations / US EPA*, 2023) (Haughton, 2024)

Through comprehensive research, we have determined that High-Performance Computing (HPC) systems utilize a combination of Central Processing Units (CPUs) and Graphics Processing Units (GPUs), each composed of several materials that collectively contribute to the overall carbon emissions associated with their production. Below is a detailed breakdown of the materials used in the manufacturing of CPUs and GPUs, the typical quantities of these materials within HPC systems, their respective emission factors, and the calculated carbon emissions for each material. (Emission Factor And Inventory Group, 1995)

CPU Materials and Emissions are shown in table 1 (Lisa, 2023)

Material	Amount Used (tons)	Emission Factor (tCO2e/ton)	Carbon Emissions (tCO2e)
Silicon	0.5	1.9	0.95
Copper	0.1	4.1	0.41
Aluminum	0.05	8.1	0.405
Rare Earth Metals	0.02	3.5	0.07

Table 3.1.0.1

GPU Materials and Emissions are shown in table 2 (Reemeyer, 2016)

Material	Amount Used (tons)	Emission Factor (tCO ₂ e/ton)	Carbon Emissions (tCO ₂ e)
Silicon	0.6	1.9	1.14
Copper	0.15	4.1	0.615
Aluminum	0.08	8.1	0.648
Rare Earth Metals	0.03	3.5	0.105

Table 3.1.0.2

After aggregating the data, we can get the total carbon emissions used in CPU and GPU respectively.

$$C_{\text{mat CPU}} = C_{\text{mat silicon}} + C_{\text{mat copper}} + C_{\text{mat aluminum}} + C_{\text{mat rare-earth}}$$

$$C_{\text{mat CPU}} = 0.95 + 0.41 + 0.405 + 0.07 = 1.835 \text{ tCO}_2\text{e}$$

For each GPU, the total carbon emission is:

$$C_{\text{mat GPU}} = C_{\text{mat silicon}} + C_{\text{mat copper}} + C_{\text{mat aluminum}} + C_{\text{mat rare-earth}}$$

$$C_{\text{mat GPU}} = 1.14 + 0.615 + 0.648 + 0.105 = 2.508 \text{ tCO}_2\text{e}$$

To sum up this part, the grand total carbon emission for each HPC system is $4.343 \text{ tCO}_2\text{e}$. This analysis illustrates the significant carbon footprint associated with the materials used in high-performance computing hardware, emphasizing the importance of considering sustainable practices in their production and use.

Transport Emission Model

The Carbon Emissions during the **transportation process** of HPC can be intuitively expressed in the form below:

$$C_{\text{trans}} = D_{\text{trans}} \cdot M_{\text{raw } i} \cdot ef_{\text{trans}}$$

Where C_{trans} refers to the total emissions during transportation of HPC's raw material to the manufacturing facility and the finished product to market. D_{trans} refers to the distance travelled. $M_{\text{raw } i}$ refers to the mass of these raw materials. ef_{trans} refers to emission factor of transport, which CO_2e emissions per kilogram per kilometer and it depends on the mode to transport, such as plane, ship, truck.

Distance. For the distance travelled D_{trans} , it can be divided into two parts: transportation of raw

materials to manufacturing facility and transportation of finish products to market. The distance for transporting raw materials to manufacturing facility is typically around **1000 km**. This figure represents the distance from the suppliers to the manufacturing plant, which is often calculated based on geographical data and logistics scenarios. (HPC for Energy Innovation, 2021) For the distribution of finished products, such as HPC chargers, the average transport distance is approximately **300 km** from the distribution centers to installation sites. (GridX, n.d.) Therefore, we can conclude the average overall distance traveled for HPC transportation is:

$$D_{\text{trans}} = 1000 + 300 = 1300 \text{ km}$$

Mass. There are three main types of HPCs, each of their mass varies greatly.

A single-GPU high-performance computer, such as NVIDIA A100 and AMD Instinct MI100, weighs about 13.6kg. (CP North America, n.d.) This is because single GPU HPCs' aim to design is to combine efficiency and portability together. Portability is an important factor for them, therefore the weight is intentionally optimized.

All **multi-GPU and CPU HPCs'** weight varies greatly and it depends on the number of nodes. Many multi-GPU HPC nodes have 2 to 8 GPUs per node, with some advanced systems featuring even more. (CSC, n.d.) (For this section, we take an average, which is 5 GPUs per node). The weight of GPU is between 0.18kg and 1.5kg, depending on the quality. High-End GPUs can weigh up to 1.5 kg, because of the cooling solutions and materials (copper, aluminum, silicon) used in their construction. Mid-Range GPUs typically weigh around 0.6 kg for many models. Budget GPUs' weight can be low to 0.18 kg for entry-level models. (Reddit, 2017) The number of nodes of different

HPCs are shown in table 3: (HRI, n.d.; Microsoft, n.d.)

Type of HPC	Number of nodes
Small, educational	50
Medium, common technology advances	1,000
Large, research institutions and colleges (extensive computational tasks)	10,000
Supercomputers (large parallel processing tasks)	100,000

Table 3.2.0.1

After that, we can compute the mass of multi-GPU HPC with the model shown below:

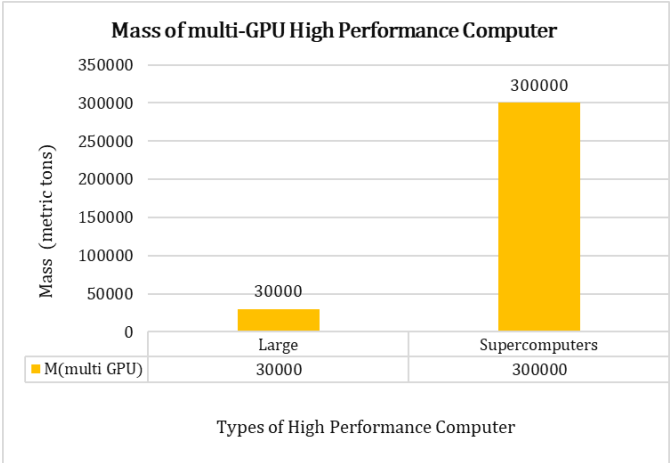
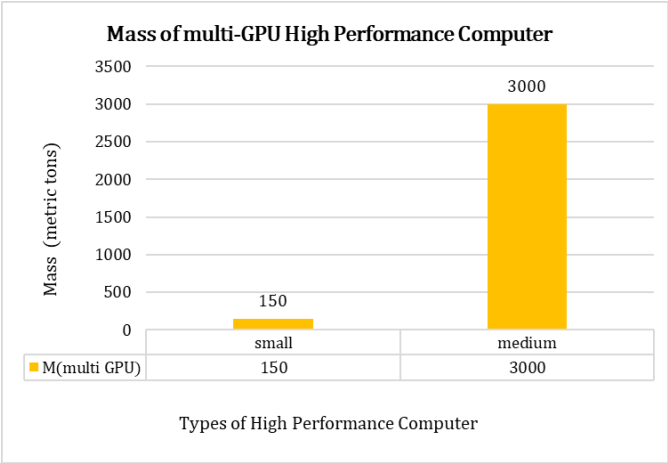
$$M_{\text{multi GPU}} = M_{\text{single GPU}} \cdot n_{\text{GPU per node}} \cdot n_{\text{node}}$$

Where $M_{\text{multi GPU}}$ refers to the mass of multi-GPU HPC computer, $M_{\text{single GPU}}$ refers to the mass of a single GPU computer, $n_{\text{GPU per node}}$ refers to the

number of GPUs on one node, which we will take the average and consider $\bar{n}_{\text{GPU per node}}$ as 5 nodes, and n_{node} refers to the number of nodes in HPC. In table 4, data are listed and the mass is calculated using the formula above, different HPC's masses are depicted as in graphs 1 and 2.

Computer types	n(nodes)	n(GPU nodes)	per	M(single GPU) (in metric tons)	M(multi GPU) (in metric tons)
small	50	5		0.0006	0.15
medium	1000	5		0.0006	3
Large	10000	5		0.0006	30
Super-computers	100000	5		0.0006	300

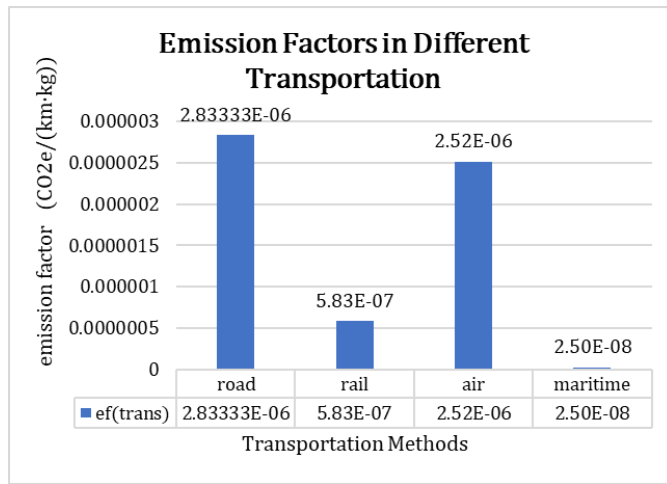
Table 3.2.0.2



Graph 3.2.0.1(2)

Emission Factor of Transport(ef_{trans})(5), refers to the average emissions of CO_2e produced per unit weight per unit length. These factors can change by types of transportation and fuel type. This table listed the emission factor for different modes of transport. Emission Factors of different transportations are shown in graph 2.

Graph 3.2.0.1(1)



Graph 3.2.0.2

We can calculate the carbon emissions of different HPCs under different methods of transportation by using this formula:

$$C_{\text{trans m c}} = D_{\text{trans m}} \cdot M_{\text{multi GPU c}} \cdot ef_{\text{trans}}$$

Where sub c stands for a type of computers, and sub m stands for a transportation method. This is a more precise version compared to the intuitive formula we developed before, with $M_{\text{multi GPU c}}$ replacing $M_{\text{raw i}}$ since raw materials are assembled in the whole transportation stage and this doesn't change the total mass.

Usage Emission Model

Carbon emission during usage can be broken down into carbon emitted by different energy sources.

$$C_{\text{usage}} = \sum_{\text{Energy Sources}} E_{\text{computation}} \cdot r_{\text{source}} \cdot k_{\text{carbon}}(\text{source})$$

Where r_{source} is the percentage of the energy source in total electricity production, and $k_{\text{carbon}}(\text{source})$ is that source's carbon emission factor (0 for renewable energy sources).

Written in vector form:

$$C_{\text{usage}} = E_{\text{computation}} \cdot \vec{r}_{\text{energy}} \cdot \vec{k}_{\text{carbon}}$$

The energy of a system consumed during a period of time can be described as:

$$E = \int P \, dt$$

Moreover, if $f_{\text{renewable}}$ represents the fraction of power usage that's from renewable sources, the

total carbon emission over time can be described as:

$$C_{\text{usage}} = \int P_{\text{computation}} \cdot \vec{r}_{\text{energy}} \cdot \vec{k}_{\text{carbon}} \cdot dt$$

Written in sum form:

$$C_{\text{usage}} = \sum_{t=t_0}^{t_1} P_{\text{computation}} \cdot \vec{r}_{\text{energy}} \cdot \vec{k}_{\text{carbon}} \cdot \Delta t$$

Energy consumption of cooling

Both power consumption (P_{cooling}) and water consumption of cooling are linearly related to the heat produced by the HPC system, which is equal to the energy consumed by the system. Data Center Dynamics mentions that only less than 0.1% of the energy consumed.

Adding the cooling factor $k_{\text{cooling energy}}$ which denotes the energy used for cooling per unit of heat emitted by the HPC system, and using a sum instead of an integral to make it computable.

$$C_{\text{usage}} = \sum_{t=t_0}^{t_1} P_{\text{computation}} \cdot (1 + k_{\text{cooling energy}}) \cdot \vec{r}_{\text{energy}} \cdot \vec{k}_{\text{carbon}} \cdot \Delta t$$

End-of-life Emission Model

As we have previously stated in '2.3 Assumptions and Justifications', carbon emissions from the destruction of HPC hardware are negligible compared to overall computing emissions.

Servers typically have a lifespan of 6 to 10 years, after which they must be properly disposed of to prevent environmental contamination and data security risks. Data sanitization methods, such as software wiping and degaussing, are carbon-neutral, while physical destruction methods, like shredding or incineration, may release toxic substances but contribute minimally to carbon emissions. Recycling and landfilling also don't release CO₂. Thus, the carbon impact of hardware destruction is insignificant and could be neglected when compared to the overall carbon footprint of HPC operations.

Apply the Model

During Material Stage

The two variables in the material stage are the amount of materials used and the emission factor for each specific material. Silicon increased from approximately 1,000 tons in 2013, 1,200 tons in 2015 to about 1,800 tons in 2020. Copper usage rose from around 600 tons in 2015 to about 800 tons in 2020. Aluminum remained stable at around 400 tons. Rare Earth Metals increased slightly from about 100 tons to 150 tons.

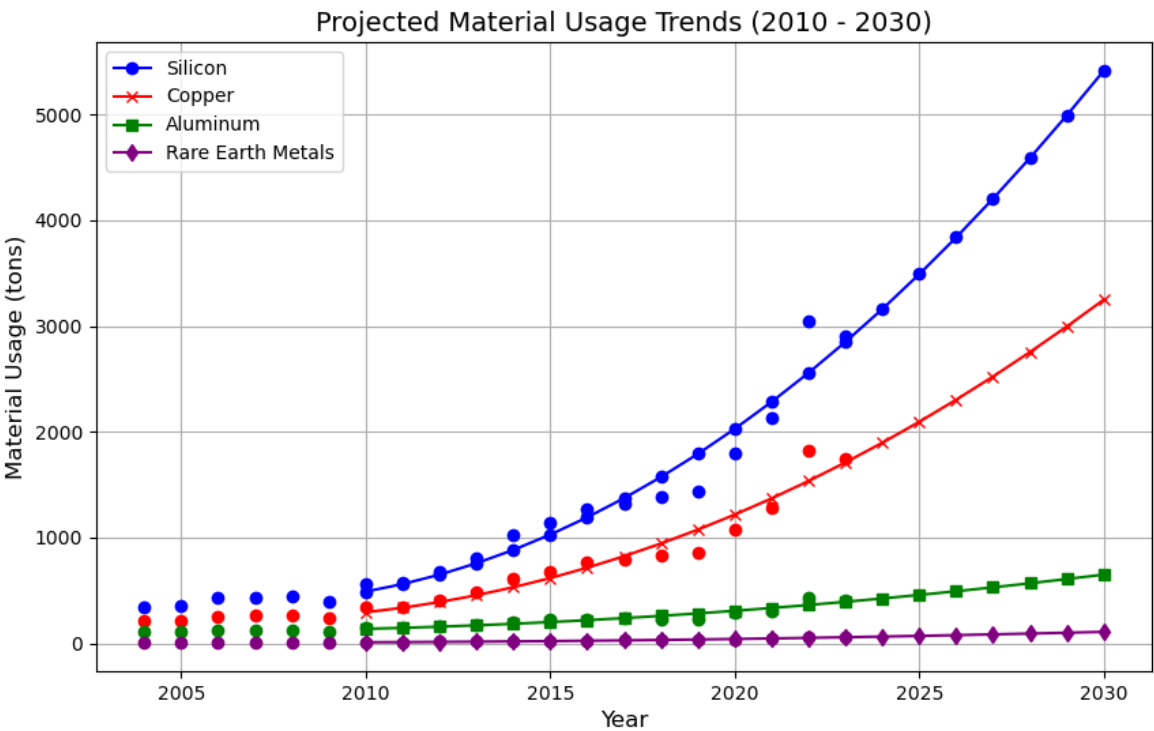
To predict the material usage trends for 2030, we will employ polynomial regression due to its ability to model non-linear relationships effectively. The following equations represent the projected usage based on historical data where y is the predicted usage for each material and x is the year.

Silicon: $y = a_1x^2 + b_1x + c_1$

Copper: $y = a_2x^2 + b_2x + c_2$

Aluminum: $y = a_3x^2 + b_3x + c_3$

Rare Earth Metals: $y = a_4x^2 + b_4x + c_4$



Graph 4.1.0.1

After gathering data and prediction, we used Python to plot the regression line where the blue regression line is silicon usage, yellow represents copper usage, green shows the aluminum usage

and red if the rare earth metal usage. <1>
In 2030, the total usage of silicon, copper, aluminum and rare materials is listed in the table below:

Material	Amount Used (tons)	Emission Factor (tCO2e/ton)	Carbon Emissions (tCO2e)
Silicon	5450	1.9	10355
Copper	3275	4.1	13427.5
Aluminum	675	8.1	5467.5
Rare Earth Metals	130	3.5	455

Table 4.1.0.1

Using this formula:

$$C_{\text{mat}} = C_{\text{mat silicon}} + C_{\text{mat copper}} + C_{\text{mat aluminum}} + C_{\text{mat-rare earth}}$$

of weight increases. Accordingly, we can estimate the carbon emissions by calculating the emissions today.

Thus, we have:

By the data mentioned above, the total weight:

$$C_{\text{mat}} = 10355 + 13427.5 + 5467.5 + 455 = 29705 \text{ } tCO_2e$$

$$M_{\text{multi GPU}} = \sum_i M_{\text{raw } i} = 5450 + 3275 + 675 + 130 = 9530 \text{ } MT$$

During Transportation Stage

$$C_{\text{trans}} = D_{\text{trans}} \cdot M_{\text{multi GPU}} \cdot ef_{\text{trans}}$$

The variables in the transportation stage are the mass of raw materials, the distance travelled and the weight of single computer, all not likely to alter much in the next 6 years. However, more hardware will be built, hence the total amount

This will be useful afterwards. Now we want to know the percentage of each combination of different transportation methods and different types of computers.

By research:

Small	Medium	Large	Super
20%	40%	30%	10%

Table 4.2.0.1

Roads	Rails	Air	Maritime
55%	25%	15%	5%

Table 4.2.0.2

Calculating the percentage by the product rule: besides roads are unrealistic, so the percentage is (Transporting supercomputer by other methods 0%)

Percentage	Roads	Rails	Air	Maritime
Small	20%×55%=11%	20%×25%=5%	20%×15%=3%	20%×5%=1%
Medium	40%×55%=22%	40%×25%=10%	40%×15%=6%	40%×5%=2%
Large	30%×55%=16.5%	30%×25%=7.5%	30%×15%=4.5%	30%×5%=1.5%
Super	10%×100%=10%	0%	0%	0%

Table 4.2.0.3

Then, we can break down the above-mentioned formula:

$$C_{\text{trans}} = D_{\text{trans}} \cdot M_{\text{multi GPU}} \cdot ef_{\text{trans}}$$
$$= \sum_{c,m} (D_{\text{trans}} \cdot M_{\text{multi GPU } c} \cdot ef_{\text{trans } m})$$
$$= \sum_{c,m} (D_{\text{trans}} \cdot M_{\text{multi GPU}} \cdot ef_{\text{trans } m} \cdot \text{percentage of different c \& m combinations})$$

Where c stands for a type of computer, and m stands for a transportation method.

Roads	1300	9530	11%	3.861233791	
	1300	9530	22%	7.722467581	
	1300	9530	16.5%	5.791850686	
	1300	9530	10%	3.510212537	
Rails	1300	9530	5%	0.361345627	
	1300	9530	10%	0.722691254	
	1300	9530	7.5%	0.54201844	
Air	1300	9530	3%	0.935370739	
	1300	9530	6%	1.870741478	
	1300	9530	4.5%	1.403056108	
Maritime	1300	9530	1%	0.00309725	
	1300	9530	2%	0.0061945	
	1300	9530	1.5%	0.004645875	
Sum	/	/	/	100%	26.73492587
Unit	km	MT	/		1000 MT
Sum in tons	/	/	/	/	26734.92587

Table 4.2.0.4

The unit is 1000 tons since:

$km \cdot MT \cdot MT \cdot km^{-1} \cdot kg^{-1} = 1000 \cdot km \cdot kg \cdot MT \cdot km^{-1} \cdot kg^{-1} = 1000 \text{ MT}$

In metric tons, the result is 26734.92587 MT

During Usage Stage

Finding $\vec{k}_{\text{carbon}}$

Since coal and natural gas are the 2 main sources of electricity that emit CO₂, $\vec{k}_{\text{carbon}}$ is broken down into 3 dimensions: coal, natural gas, and renewable energies.

Energy source	produced per unit volume/mass of fuel burned	Electrical energy produced per unit volume/-mass of fuel	produced per watt-hour of electrical energy generated using source (kg/kWh)
Coal	1735.85 kg/short ton (EIA, 2024)	1.14 pounds/kWh (<i>Frequently Asked Questions</i> , n.d.)	0.989
Natural gas	54.81 kg/1000 cubic feet (U.S. Energy Information Administration, 2023)	7.42 cubic feet/kWh (<i>Frequently Asked Questions</i> , n.d.)	0.407

Table 4.3.1.1

Energy sources and the rate they produce CO₂

Carbon emission caused by renewable energies is considered negligible compared to coal and natural gas, so $k_{\text{renewable}} = 0$ is used.

$$\vec{k}_{\text{carbon}} = \begin{pmatrix} k_{\text{coal}} \\ k_{\text{natural gas}} \\ k_{\text{renewable}} \end{pmatrix} = \begin{pmatrix} 0.989 \\ 0.407 \\ 0 \end{pmatrix}$$

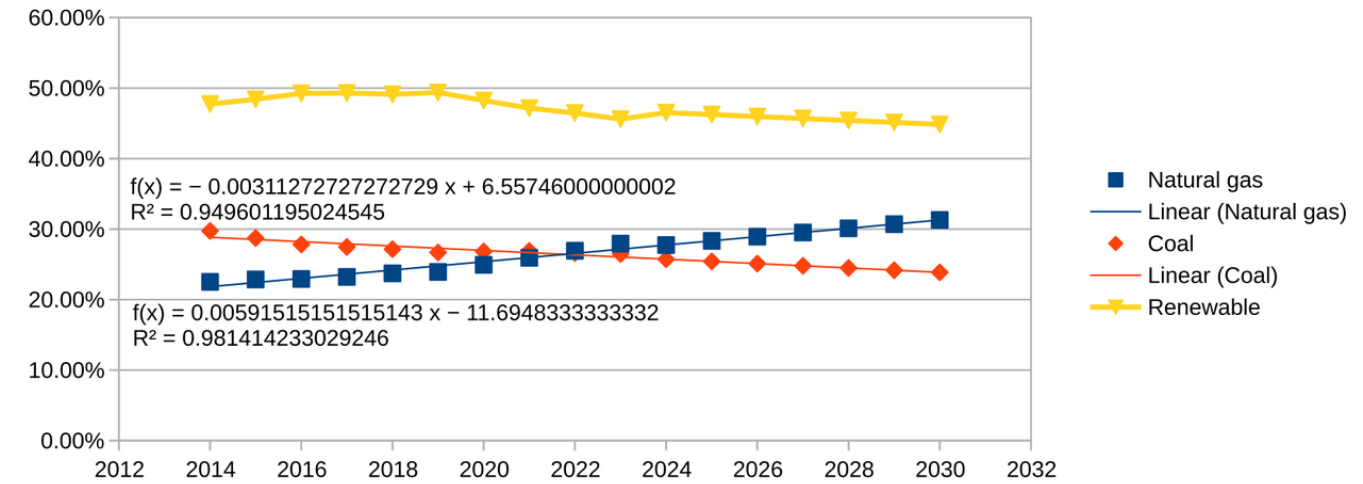
Predicting $\vec{r}_{\text{energy}}$ in the future

$\vec{r}_{\text{carbon}}$ can be defined similarly:

$$\vec{r}_{\text{carbon}} = \begin{pmatrix} r_{\text{coal}} \\ r_{\text{natural gas}} \\ r_{\text{renewable}} \end{pmatrix}$$

Using data of percentage from (*Energy consumption by source*, n.d.), a prediction of the percentage of electricity generated by natural gas and coal, is made below. In this case, $r_{\text{renewable}}$ can be calculated by:

$$r_{\text{renewable}} = 100\% - r_{\text{coal}} - r_{\text{natural gas}}$$



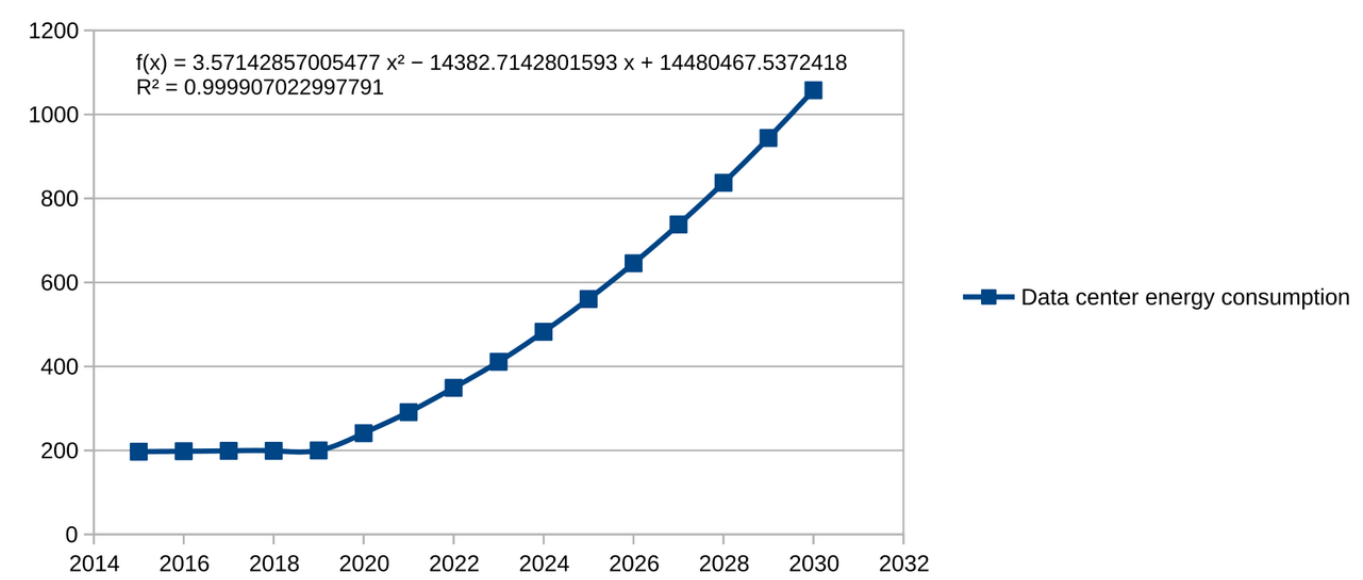
Graph 4.3.2.1

Year	(%)	(%)	Year	(%)	(%)
2014	22.51	29.75	2023	27.93	26.47
2015	22.86	28.73	2024	27.74	25.73
2016	22.93	27.82	2025	28.33	25.42
2017	23.22	27.47	2026	28.93	25.11
2018	23.73	27.15	2027	29.52	24.8
2019	23.93	26.69	2028	30.11	24.48
2020	24.93	26.83	2029	30.7	24.17
2021	25.93	26.91	2030	31.29	23.86
2022	26.93	26.6	/	/	/

Table 4.3.2.1 Describes the trend in the ratio of electricity generated from different sources.

Modelling $E_{\text{computation}}$

Using a degree-2 polynomial regression on the annual energy consumption of data centers from 2019 to 2023, below is a table and a graph of the predicted annual energy consumption of data centers from 2024 to 2030.



Graph 4.3.3.1

Year	Data center energy consumption (TWh)	Year	Data center energy consumption (TWh)
2015	197	2023	411
2016	198	2024	482
2017	199	2025	560
2018	199	2026	646
2019	200	2027	738
2020	241	2028	837
2021	291	2029	944
2022	349	2030	1060

Table 4.3.3.1

The trend in the global electrical energy consumption of data centers.

Predicting C_{usage}

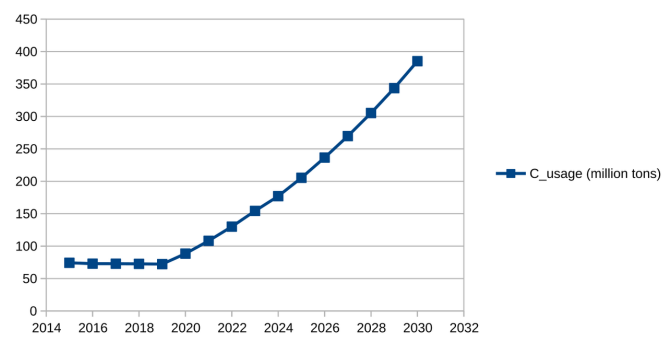
When modeling the carbon emission each year and using $\Delta t = 1$ year, the equation

$$C_{\text{usage}} = \sum_{t=t_0}^{t_1} P_{\text{computation}} \cdot (1 + k_{\text{cooling energy}}) \cdot \vec{r}_{\text{energy}} \cdot \vec{k}_{\text{carbon}} \cdot \Delta t$$

can be simplified to

$$C_{\text{usage}}(t) = \frac{E_{\text{computation with cooling}}(t)}{\vec{k}_{\text{carbon}}} \cdot \vec{r}_{\text{energy}}(t)$$

The result of C_{usage} is shown in the table and graph below



Graph 4.3.4.1

Year	(million tons)	Year	(million tons)
2015	74.3	2023	154
2016	73	2024	177
2017	72.9	2025	205

Year	(million tons)	Year	(million tons)
2018	72.7	2026	236
2019	72.3	2027	270
2020	88.4	2028	305
2021	108	2029	344
2022	130	2030	385

Table 4.3.4.1

Describes the trend in the global carbon dioxide emission from data centers by electricity consumption.

Considering both table and graph, the predicted quantified annual CO₂ emission in 2030 is 385 million metric tons.

During End-of-Life Stage

Carbon emissions are still negligible. Even if there are changes, it decreases in carbon emissions. This is due to fact that people are constantly searching resolutions to mitigate carbon emissions. Thus, we conclude its omissability.

Result of the Model

By the formula:

$$C_{\text{total}} = \sum C = C_{\text{mat}} + C_{\text{trans}} + C_{\text{usage}} + C_{\text{end}}$$

We are able to determine the total emission in most years. In 2030, the annual CO₂ is:

$$\begin{aligned} C_{\text{total}} &= \sum C = 29,705 + 26,734.92587 + \\ &385,000,000 = 385,056,439.92587 \text{ MT} \\ &= 385.05644 \text{ MMT} \end{aligned}$$

Yet merely considering the carbon emissions while running HPCs, concluded also in '4.3':

$$C_{\text{usage}} = 385 \text{ MMT} = 385000000 \text{ MT}$$

Refine Previous Model

Water Usage

Water usage is chosen as the second impact to be modeled. Here, we have several reasons why we chose this topic. One of them is that realizing how precious water is, we eagerly wanted to figure out approximately how much water was used or consumed. Another is just simply that nearly all

water consumption comes from evaporative cooling of the computing components, it is somehow easier to quantify.

Water usage relates to other fields such as land use and socioeconomic impacts. Considering the optimum place for constructing data centers requires pondering on the amount of water that will probably be used and whether removing huge amount of water from nearby environments contribute to landsliding, droughts or other disasters.

Similar to energy consumption of cooling, water consumption has an approximately linear relationship to the energy consumption of HPC systems during their usage, which is already modeled above.

Let $k_{\text{cooling water}}$ denote water used for cooling per unit of heat emitted by the HPC system, water consumption during usage can be modeled with the equation below.

$$\text{Water}_{\text{total}} = \sum_{t=t_0}^{t_1} P_{\text{Computation}} \cdot k_{\text{cooling water}} \cdot \Delta t$$

Thus, tracking the water usage of an HPC system also reflects how much energy it consumes.

CPU, GPU, and RAM Power Consumption Trends

Hardware like CPU, GPU, and RAM are becoming more power efficient in recent years, however, not all HPC systems will immediately start to use the newest generation of chips once they are released, so the average power consumption of each type of chip should be a moving average of the power consumption of the newest generation of chips. Assume that a newer generation of chips will take time t_r to fully replace the older generation:

$$P_{\text{average}}(t) = \frac{1}{t_r} \cdot \int_{t-t_r}^t P_{\text{newest}}(t) dt$$

Apply Refined Model

Future trends

How model changes if HPC grows

This refined model determines the total amount of water used to cool down HPCs. The result relates to the water usage of different HPC parts, including CPU, GPU, RAM and the infrastructure. It also relates to the water used to cool down per unit of heat and time. The numbers of CPU, GPU, RAM and infrastructure are directly related to the number of HPCs. Therefore, with the growth of HPCs, the number of the parts also increases. So, the $\text{Water}_{\text{total}}$ will increase with the growth of HPCs.

How model changes if energy demanded in other sectors grows

If energy demanded in other HPC sectors grows in the future, the heat generated as energy lost will also increase. Water usage is directly proportional to the amount of heat in order to keep the temperature stable. Thus, water usage, also known as $\text{Water}_{\text{total}}$, increases with the growth of energy demanded in other sectors of HPC.

How model changes if energy sources change

Water usage is related to heat. Recently, more non-renewable energy resources are used to generate energy for HPC. Considering the environmental impact, people don't generate excess energy---they only generate energy that is right enough for usage. In the future, major energy resources will shift from non-renewable ones to renewable ones, which means there are no environmental-protection limits to generate energy. People will generate excess amounts of energy to ensure there is enough energy for HPC usage. Therefore, $\text{Water}_{\text{total}}$ will increase if energy resources shift.

Predicting $\text{Water}_{\text{total}}$ in future

1 cubic meter (1000 L) of water can produce up to 695 kWh of cooling power (*Use Water to Save Water*, 2023), or

$$k_{\text{cooling water}} = \frac{1000L}{695kWh} = 1.44L/kWh$$

1 kWh of electricity used for evaporative cooling can provide up to 40kWh of cooling power (*Evaporative cooling*, n.d.), or

$$k_{\text{cooling energy}} = \frac{1kWh}{40kWh} = 0.025$$

Using the equation

$$C_{\text{usage}} = \sum_{t=t_0}^{t_1} P_{\text{computation}} \cdot (1 + k_{\text{cooling energy}}) \cdot \vec{r}_{\text{energy}} \cdot \vec{k}_{\text{carbon}} \cdot \Delta t,$$

and using $\Delta t = 1$ year,

$$E_{\text{computation}} \cdot (1 + k_{\text{cooling energy}}) = E_{\text{computation and cooling}}$$

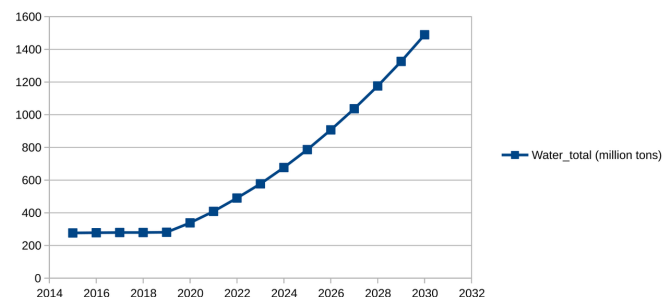
$$E_{\text{computation}} = \frac{E_{\text{computation and cooling}}}{1 + k_{\text{cooling energy}}}$$

can be calculated for each year.

Then using

$$\begin{aligned} \text{Water}_{\text{total}} &= \sum_{t=t_0}^{t_1} P_{\text{computation}} \cdot k_{\text{cooling water}} \cdot \Delta t \\ &= E_{\text{computation}} \cdot k_{\text{cooling water}}, \end{aligned}$$

the result of $\text{Water}_{\text{total}}$ is shown in the chart and table below.



Graph 6.2.0.1

Year	Water_total (MMT)	Year	Water_total (MT)
2015	277	2023	577
2016	278	2024	677
2017	280	2025	787
2018	280	2026	908
2019	281	2027	1040
2020	339	2028	1180
2021	409	2029	1330
2022	490	2030	1490

Table 6.2.0.1

The table shows the trend in total water consumed by data centers through evaporative cooling. According to the table, the figure is 1490 MMT.

Impact of replacing old hardware faster

Using the refined model, during usage,

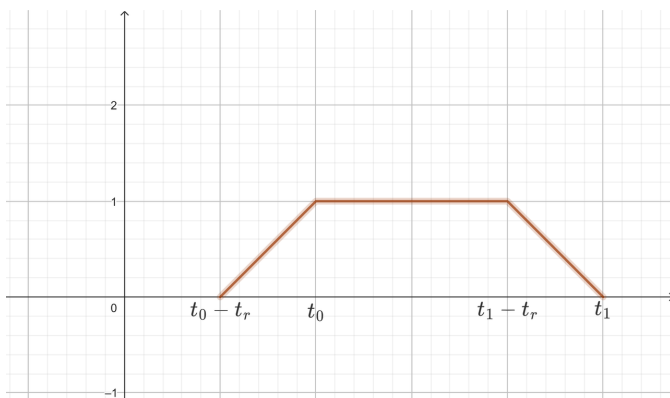
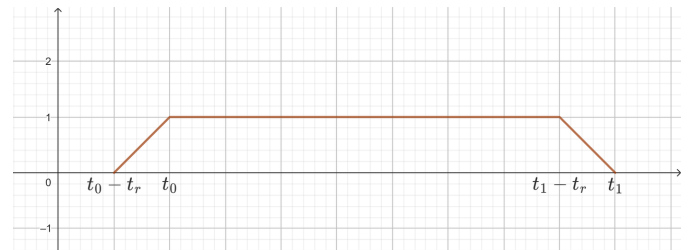
$$E_{\text{total}} = \int_{t_0}^{t_1} P_{\text{average}}(t) dt \quad (1)$$

$$= \int_{t_0}^{t_1} \frac{1}{t_r} \cdot \int_{t-t_r}^t P_{\text{newest}}(t) dt dt \quad (2)$$

$$= \int_{t_0}^{t_1} P_{\text{newest}}(t) \cdot W(t) dt \quad (3)$$

(The last step is done by calculating the integral as the limit of a double sum, simplify using summation techniques, then finding the piecewise expression of $W(t)$ as Δt approaches 0.)

Where $W(t)$ passes through $(t_0 - t_r, 0)$, $(t_0, 1)$, $(t_1 - t_r, 1)$, and $(t_1, 0)$, and has straight lines between these points. The graph of $W(t)$ is shown below.

**Graph 6.3.0.1****Graph 6.3.0.2**

Notice that no matter the value of t_r , the value of $W(t)$ when $t_0 \leq t \leq t_1 - t_r$ is always 1. As $t_1 - t_0$ gets larger and t_r becomes negligible relative to $t_1 - t_0$, the change in $W(t)$ where t is in the interval $[t_0 - t_r, t_0] \cup [t_1 - t_r, t_1]$ has negligible effects on the value of E_{total} .

Thus, even if replacing old hardware with newer and more efficient hardware frequently will save power in the short term, the amount of energy saved by doing so does not increase over time, and the effect will become negligible in the long term. On the other hand, frequently changing hardware creates a large amount of e-waste, which constantly creates negative environmental impacts from leaking chemicals and energy consumption for disposal.

Strengths and Weaknesses

Strengths

First, analysing, we divided the life cycle of HPC systems into four major stages, material obtaining and manufacturing, transportation, computation, and end-of-life. This model developing method helps establish a granular understanding of where environmental impacts arise. This clear and effec-

tive segmentation allows us to target interventions at different stages (e.g., reducing energy consumption during the computational stage or focusing on recycling materials at end-of-life), which is a robust approach for sustainability analysis. This guarantees a more precise policy recommendations and technical improvements at each stage of the HPC system's life cycle.

Second, when designing energy emission model during usage, we effectively incorporate a moving average for power consumption trends of CPUs, GPUs, and RAM, based on the time-to-replacement cycle for newer chip generations. This approach allows our model to stand out from other competitors' or previous calculations and models, accounting for transitions between hardware generations, where the newest chip architectures might not immediately replace older generations in large-scale HPC deployments. By integrating a time-based averaging mechanism, our approach provides a more accurate representation of power consumption over time, resulting in a more realistic estimate of energy demand throughout the lifespan of the system.

Third, our model includes **regression-based predictions** for future computing demand, both in terms of computational power (FLOPS) and memory usage. This makes the model adaptable to future trends in AI and HPC infrastructure, ensuring that predictions for carbon emissions and water usage remain relevant as demand grows. Additionally, the explicit inclusion of scaling factors like P_{CPU} , P_{GPU} , P_{RAM} per FLOPS and per memory capacity provides a good mechanism for incorporating growth in system scale into the calculations.

Fourth, by incorporating the mass of different types of HPCs (single-GPU vs. multi-GPU) and specifying their weight and node configurations, our transportational model makes **accurate allowances** for the varying sizes of HPC systems, which enhances the **flexibility** of the model, allowing it to scale across different kinds of deployments. The **adjustable parameters** ensure the model can be used for both **theoretical studies** and practical real-world implementations.

Weaknesses

While our model has many strengths, we still have some constraintss.

First, we need to assume carbon emissions from hardware destruction is close to zero, so the result may be flawed.

Second, our model assumes a steady, linear progression in technological advancements with respect to power consumption per FLOP (floating-point operation). However, this assumption may not fully capture the complexity of real-world developments in chip design, particularly with the introduction of new hardware architectures or unexpected shifts in manufacturing technologies (e.g., **3D stacking** of chips or **quantum computing** advancements). Moreover, using a moving average to smooth power consumption trends may not fully reflect the potential for **disruptive innovations** that could significantly reduce power consumption or improve energy efficiency in ways that are not captured by historical regression models.

Additionally, the use of a moving average to smooth power consumption trends might fail to account for disruptive innovations that could drastically reduce power consumption or improve energy efficiency in ways that are not captured by historical regression models. This heavy reliance on regression assumes that the higher-order derivatives of most variables over time are either zero or other constant, which may not hold in the face of rapid technological change.

Recommendations to reduce environmental impact

HPCs do bring lots of convenience to our life, such as much quicker computing. However, the usage will emit carbon compounds, which is known as carbon emissions. Carbon emissions will bring a myriad of environmental problems. Its compound, carbon dioxide, a major kind of greenhouse gas, has outstanding capability to lock the heat and constantly increase the global temperature. This result in some environmental disasters, such as glacier melting, sealevel rising, which eventually leads to unbalance of some ecosystems. Lots of species will be endangered or extinct, spaces for human beings will decrease, etc.

Also, great amount of water is used for cooling purpose and it's wasted after usage. Water is scarce on earth, constantly wasting water can also lead to environmental problems. People should have a fixed inclination towards devising relative

remedies in order to greatly reduce environmental impacts. Some actions must be done, our recommendations are elucidated below.

Hardware Optimization

An increase in hardware efficiency can reduce carbon emissions and water usage. So new generations of hardware should be sought constantly. To encourage the invention and investigation of prior hardware, government can introduce encouragement policies, such as giving welfare to companies which investigate new hardware. If the policy become true in the future, citizens are motivated to invent new generations of hardware. However, there will be a time period after releasing old hardware and before using the new one. According to NYU, many HPC facilities, such as those at NYU, typically retire equipment after four years of service [25]. According to the university of Iowa, its model allows for a maximum lifespan of seven years for hardware, such as compute nodes, with a warranty covering the first five years. Therefore, we can conclude the average time begin to replace is 4-5 years while the time used to fully replace is 7 years. Using the model above:

$$P_{\text{average}}(t) = \frac{1}{t_r} \cdot \int_{t-t_r}^t P_{\text{newest}}(t) dt$$

Where t is given as 7 years, t_r as 4.5 years. Calculating, the average power used in the 7 years, which includes the use of old generation, the use of both until the fully use of new generation is:

$$P_{\text{average}}(t) = \frac{1}{4.5} \cdot \int_{2.5}^7 P_{\text{newest}}(t) dt$$

In conclusion to this recommendation, if the time used to begin refreshing t_r fluctuates and the time to fully replace t is also short, the power used can significantly change by a percentage. However, if the time takes to fully replace t is long, the change of time used to begin refreshing (change in t_r) will not alter the result seriously.

Software Improvements

In order to improve energy efficiency, software enhancements are as important as hardware optimization. Better computing methods can do more algorithms in using same amount of energy.

Also compilers impacts the use of energy. For example, we can use compiled languages instead of interpreted languages because compiled languages are more energy-efficient[26]. So by using the formula

$$E = \int P dt$$

Where E refers to energy used, P refers to power and t refers to time. A decrease in the time used to compute can decrease the total energy use.

Waste Heat Reuse and Water Reuse

The operation of HPCs can generate great amount of heat. Though some heat is cooled down with water, there is still some heat left. Also, constantly using water can cause water waste, which also has a huge impact. Therefore, if some systems are implemented to capture waste heat and convert them to energy for reuse, the energy usage percentage can also increase. According to [27], the LUMI supercomputer in Finland plans to utilize its waste heat to warm local buildings, achieving a zero or negative carbon footprint by using 100% renewable energy.

The water used for cooling purposes can actually not be wasted. In fact, it can be reused, too. According to NREL, "it would consume approximately 2 million gallons of water annually to support cooling of the IT load—approaching an hourly average of 1 megawatt[28]." Huge amount of water is used to cool down HPCs every year. If government can sponsor to establish stations to collect the water and use tubes to transport them to waterworks. Then waterworks manufacture it to the water people use everyday. This can make the wasted water to be fully reused.

Geographical Flexibility

To generate energy for HPC, using renewable energy resources is always prior to using non-renewable energy resources. If HPCs are placed in areas abundant with renewable energy resources, more of them can be utilized. Also, having nearby renewable energy resources can minimize the time needed for transportation and processing. By applying the model above:

$$C_{\text{usage}} = \int P_{\text{computation}} \cdot \vec{r}_{\text{energy}} \cdot \vec{k}_{\text{carbon}} \cdot dt$$

This model shows if the fraction of renewable energy resources increase, carbon emissions decrease.

Conclusions

In this study, our group developed a comprehensive model to estimate the carbon emissions and water usage associated with HPC systems across various stages of their lifecycle, from material acquisition, transportation to during usage and end-of-life. By considering key variables such as material emissions, methods of transport, transport emissions, energy consumption, and hardware usage, we derived formulas that calculate the total emissions for any given year. Our model also incorporated refined factors such as water usage for cooling, power consumption trends of CPU, GPU, and RAM components, and the impact of hardware replacement cycles.

Our work has several strengths which help us to stand out of other competitor models, including the use of a moving average to account for power consumption trends across hardware generations, regression-based predictions for future computing demand, and the ability to scale across different HPC configurations. These features make the model adaptable and relevant for long-term sustainability analysis.

The results of our model suggest that, in 2030, the total carbon emissions from HPC will amount to approximately 385.05643 MMT of CO₂, with materials contributing to 29705 MT, transportation accounting for 26734.925 MT, and 385 MMT during usage, with the end-of-life emissions being omissible. Furthermore, our refined model demonstrated the critical role that water usage plays in sustaining HPC systems, with predictions indicating a rise in water consumption due to the growing demands of HPC infrastructures, increased energy use in other sectors, and shifts toward renewable energy sources.

As we look to the future, our model predicts the total water usage for cooling will increase proportionally due to the rising number of components like CPUs, GPUs, and RAM. Additionally, if energy demand in other sectors of HPC rises, the

associated heat generation will also increase, leading to higher water usage for cooling purposes. Lastly, a shift from non-renewable to renewable energy sources will likely lead to increased water usage, as excess energy will be generated to ensure sufficient supply for HPC, further driving the need for cooling systems.

In a nutshell, while our model provides valuable insights into the environmental impact of HPC, future efforts should focus on continued development of energy-efficient hardware, software optimization, and innovative cooling methods based on our findings. By addressing these areas, we can help ensure that the growth of HPC systems aligns with global sustainability goals.

References

- [1] Li, B., Roy, R. B., Wang, D., Samsi, S., Gadeppally, V., & Tiwari, D. (2023). Toward Sustainable HPC: Carbon Footprint Estimation and Environmental Implications of HPC Systems. *Toward Sustainable HPC: Carbon Footprint Estimation and Environmental Implications of HPC Systems*. <https://doi.org/10.1145/3581784.3607035>
- [2] McInnes, L. C., Heroux, M. A., Draeger, E. W., Siegel, A., Coghlan, S., & Antypas, K. (2021). How community software ecosystems can unlock the potential of exascale computing. *Nature Computational Science*, 1(2), 92–94. <https://doi.org/10.1038/s43588-021-00033-y>
- [3] Lawrence Livermore National Laboratory. (2021, May 13). *HPC for energy innovation: Manufacturing brochure*. https://hpc4energyinnovation.llnl.gov/sites/hpc4energyinnovation/files/2022-03/HPC_Manufacturing_Brochure_MAY_13_2021.pdf
- [4] GridX. (n.d.). *High power charging*. <https://www.gridx.ai/knowledge/high-power-charging>
- [5] CP North America. (n.d.). *What is high performance computing and why is it critical for defense & military applications?* <https://cpnorthamerica.com/whitepapers/what-is-high-performance-computing-and-why-is-it-critical-for-defense-military-applications/>
- [6] CSC - IT Center for Science. (n.d.). *Tutorial: Multi-node parallel machine learning with CSC resources*. <https://docs.csc.fi/support/tutorials/ml-multi/>

- [7] PCMasterRace. (2017, March 18). *What does a modern GPU weigh?* Reddit. https://www.reddit.com/r/pcmasterrace/comments/56me6e/what_does_a_modern_gpu_weigh/
- [8] HRI (Homi Bhabha Research Institute). (n.d.). *Cluster status*. <https://www.hri.res.in/cluster/clusterStatus/cluster.php>
- [9] Microsoft. (n.d.). *Understanding node metrics and properties in HPC Cluster Manager*. Microsoft Docs. <https://learn.microsoft.com/ja-jp/powershell/high-performance-computing/understanding-node-metrics-and-properties-in-hpc-cluster-manager?view=hpc19-ps>
- [10] *Calculating emissions | Minnesota Pollution Control Agency*. (n.d.). Minnesota Pollution Control Agency. <https://www.pca.state.mn.us/business-with-us/calculating-emissions>
- [11] Emission Factor And Inventory Group. (1995). *Emission Factors and Emission Inventories*. <https://www3.epa.gov/ttnchie1/ap42/c00s00.pdf>
- [12] General Conformity Training Modules: Appendix A Sample Emissions Calculations | US EPA. (2023, December 21). US EPA. <https://www.epa.gov/general-conformity/general-conformity-training-modules-appendix-sample-emissions-calculations>
- [13] Houghton, N. (2024, April 25). *What Is The Carbon Footprint Of A Laptop?* Circular Computing. <https://circularcomputing.com/news/carbon-footprint-laptop/>
- [14] Lannelongue, L. (2023, October 11). *Carbon footprint, the (not so) hidden cost of high performance computing*. BCS. <https://www.bcs.org/articles-opinion-and-research/carbon-footprint-the-not-so-hidden-cost-of-high-performance-computing/>
- [15] Lisa, H. (2023, September 1). *Calculation of the Product Carbon Footprint*. HPC. <https://hpc.de/reader/calculation-of-the-product-carbon-footprint.html>
- [16] Reemeyer, L. (2016, November 24). *What drives GHG emissions from copper production?* — Resourceful Paths. Resourceful Paths. <https://www.resourcefulpaths.com/blog/2016/11/23/what-drives-ghg-emissions-from-copper-production>
- [17] *Understand your copper emissions*. (n.d.). <https://www.carbonchain.com/blog/understand-your-copper-emissions>
- [18] Judge, P. (2024, February 13). *Combining heat and compute*. Data Center Dynamics. <https://www.datacenterdynamics.com/en/analysis/combining-heat-and-compute/>
- [19] AMO. (2021). *High performance computing for manufacturing*. U.S. Department of Energy. https://hpc4energyinnovation.llnl.gov/sites/hpc4energyinnovation/files/2022-03/HPC_Manufacturing_Brochure_MAY_13_2021.pdf
- [20] Reddit. (2017, February 20). *What does a modern GPU weigh?* Reddit. https://www.reddit.com/r/pcmasterrace/comments/56me6e/what_does_a_modern_gpu_weigh/
- [21] Greenhouse Gas Protocol Initiative. (2022). *Chapter 4: Quantification of emissions from the industrial sector*. Greenhouse Gas Protocol Initiative. <https://ghgprotocol.org/sites/default/files/2022-12/Chapter4.pdf>
- [22] New Zealand Ministry for the Environment. (2022). *Corrections to the 2019 measuring emissions guide for organisations*. Ministry for the Environment. <https://environment.govt.nz/assets/Publications/Files/corrections-to-2019-measuring-emissions-guide-for-organisations.pdf>
- [23] Office of Rail and Road. (2022). *Rail and environment statistics*. Office of Rail and Road. <https://dataportal.orr.gov.uk/statistics/infrastructure-and-environment/rail-environment/>
- [24] Sinay. (2021, December 8). *How much does the shipping industry contribute to global CO2 emissions?* Sinay. <https://sinay.ai/en/how-much-does-the-shipping-industry-contribute-to-global-co2-emissions/>
- [25] *NYU High Performance Computing - HPC Policies*. (n.d.). <https://sites.google.com/nyu.edu/nyu-hpc/about/hpc-policies>
- [26] Pereira, R., Couto, M., Ribeiro, F., Rua, R., Cunha, J., Fernandes, J. P., & Saraiva, J. (2017). *Energy efficiency across programming languages: How do energy, time, and memory relate* <https://doi.org/10.1145/3136014.3136031>
- [27] *HPC at the core of green and digital transition | Scientific Computing World*. (n.d.). <https://www.scientific-computing.com/viewpoint/hpc-core-green-and-digital-transition>

- [28] *High-Performance Computing Data Center Water Usage Efficiency*. (n.d.). Computational Science | NREL. <https://www.nrel.gov/computational-science/reducing-water-usage.html>
- [29] George, R. (2024, October 30). *21 Hard Drive Data Destruction Methods: Shredding, Degaussing*. . . IT Asset Management Group. <https://www.itamg.com/data-storage/hard-drive/destruction-methods/>
- [30] Verity Systems. (n.d.). *Degaussing: what is it and how does It work?* - VS Security. <https://www.veritysystems.com/degaussing-an-introduction/>
- [31] *Green HPC: An analysis of the domain based on Top500*. (n.d.). <https://arxiv.org/html/2403.17466v1>
- [32] Andy_Patrizio. (2018, December 3). IDC: Expect 175 zettabytes of data worldwide by 2025. *Network World*. <https://www.networkworld.com/article/966746/idc-expect-175-zettabytes-of-data-worldwide-by-2025.html>
- [33] HRI. (n.d.). *Cluster Status*. Available at: <https://www.hri.res.in/cluster/clusterStatus/cluster.php> and <https://learn.microsoft.com/ja-jp/powershell/high-performance-computing/understanding-node-metrics-and-properties-in-hpc-cluster-manager?view=hpc19-ps>
- [34] *Energy consumption by source*. (n.d.). Our World in Data. <https://ourworldindata.org/grapher/energy-consumption-by-source-and-country>
- [35] U.S. Energy Information Administration. (2023, September 7). *Environment - U.S. Energy Information Administration (EIA) - U.S. Energy Information Administration (EIA)*. Eia.gov. https://www.eia.gov/environment/emissions/co2_vol_mass.php
- [36] *Frequently Asked Questions (FAQs) - U.S. Energy Information Administration (EIA)*. (n.d.). Wwww.eia.gov. <https://www.eia.gov/tools/faqs/faq.php?id=667&t=3>
- [37] Euromines. (2020). The Electronics Value Chain and Its Raw Materials. In *European Association of Mining Industries, Metal Ores & Industrial Minerals* (pp. 2–11). https://www.euromines.org/files/key_value_chain_electronics_euromines_final.pdf
- [38] Goldman Sachs. (2024, May 14). *AI is poised to drive 160% increase in data center power demand*. Wwww.goldmansachs.com. <https://www.goldmansachs.com/insights/articles/AI-poised-to-drive-160-increase-in-power-demand>
- [39] *Keys to achieving water efficiency in data centers*. (2024, June 21). Vertiv.com. <https://www.vertiv.com/en-latam/about/news-and-insights/articles/blog-posts/Keys-to-achieving-water-efficiency-in-data-centers/>
- [40] Sahai, R., Shah, N., & Phadke, A. (n.d.). *Addressing Water Consumption of Evaporative Coolers with Greywater*. <https://www.aceee.org/files/proceedings/2012/data/papers/0193-000025.pdf>
- [41] *Use Water to Save Water: The Efficiency of Indirect/Direct Evaporative Cooling*. (2023). Oxy-Com.com. <https://www.oxy-com.com/blog-news/evaporative-cooling-water-consumption>
- [42] *Evaporative cooling vs air conditioning*. (n.d.). Wwww.oxy-Com.com. <https://www.oxy-com.com/evaporative-cooling-vs-airconditioning>

Letter to the Advisory Board

Dear members of the Advisory Board,

I am writing to advocate for the inclusion of a more detailed section on the environmental impacts of High-Performance Computing (HPC) in the developmental goals for 2030. As we progress into an era where computational power plays a pivotal role in advancing scientific discovery and technological innovation, it is equally imperative that we address the associated environmental consequences. While HPC offers numerous benefits, its rapid growth also contributes to significant ecological challenges, particularly in terms of carbon emissions, energy consumption, and water usage.

Our team has been working rigorously to develop models that quantify and track the environmental footprint and water usage of HPC systems across their entire lifecycle, ranging from material and transportation to usage and end-of-life stages. As the demand for computational power continues to

rise, these environmental impacts are poised to exacerbate the broader climate crisis. Therefore, it is essential that we proactively address these challenges as part of our strategic goals for the future.

To mitigate the negative environmental consequences of HPC, we propose the following recommendations based on our findings. First, we emphasize the importance of investing in energy-efficient hardware. Modernizing HPC systems with energy-efficient components can significantly reduce both carbon emissions and water consumption. Moreover, incentivizing the research and development of next-generation hardware through government-backed initiatives can encourage companies to innovate in a more sustainable direction. For instance, current practices of retiring hardware after four to seven years contribute to inefficient power use. By optimizing the lifecycle of HPC equipment, including improving reuse and recycling strategies, we could achieve substantial reductions in overall energy consumption (World Wide Technology, n.d.; CIO, 2023).

Second, it's equally crucial to enhance software efficiency. By utilizing more efficient algorithms and compiled languages over interpreted ones, we can achieve better performance with lower energy requirements. This approach can directly decrease the total energy consumed during computations, aligning with our sustainability goals (CIO, 2023)(IBM Community, 2022). Moreover, implementing systems to capture waste heat from HPC operations can convert this energy for reuse, as demonstrated by initiatives like the LUMI supercomputer in Finland which plans to use its waste heat for local heating. Additionally, reusing water utilized for cooling estimated at around 2 million gallons annually for some facilities can alleviate water scarcity issues while promoting sustainability.

To sum up, this is not merely an option, but a necessity for responsible and sustainable technical advancement. By adopting the recommendations outlined above, this will not only enhance the long-term sustainability of our computing infrastructure, but it will also help to mitigate the ecological consequences of our collective technological progress, ensuring a more sustainable future for generations to come.

Thank you for considering this crucial aspect of our HPC strategy. We look forward to your positive response and to collaborating on the integration of these recommendations.

Sincerely,

The HIMCM Team #15733

Appendices

Codes

<1> Python

```
import numpy as np
import matplotlib.pyplot as plt

# Historical data for each material
years = np.array([2004,
2005,
2006,
2007,
2008,
2009,
2010,
2011,
2012,
2013,
2014,
2015,
2016,
2017,
2018,
2019,
2020,
2021,
2022,
2023

                ])
silicon_usage = np.array([2572,
2666,
3174,
3226,
3332,
2957,
4195,
4271,
5067,
5970,
7628,
8435,
9479,
9774,
10310,
10700,
13390,
15870,
22640,
21620,

                ]) / 7.4388
copper_usage = silicon_usage / 0.5 * 0.3
```

```

aluminum_usage_fix = np.array([
1.5,
1.5,
1.5,
1.4,
1.4,
1.4,
1.3,
1.3,
1.2,
1.1,
1.0,
1.0,
0.9,
0.9,
0.8,
0.8,
0.8,
0.7,
0.7,
0.7,
])
aluminum_usage = silicon_usage / 0.5 * aluminum_usage_fix / 10
rare_earth_usage = silicon_usage / 0.5 * 0.01

# Fit polynomial regression (degree 2) to the data
silicon_poly = np.polyfit(years, silicon_usage, 2)
copper_poly = np.polyfit(years, copper_usage, 2)
aluminum_poly = np.polyfit(years, aluminum_usage, 2)
rare_earth_poly = np.polyfit(years, rare_earth_usage, 2)

# Generate a range of years from 2010 to 2030 for prediction
future_years = np.arange(2010, 2031)

# Predict usage for each material using the poly

```

Report on the use of AI

(1) OpenAI *ChatGPT* (Nov. 7, 2024 version, ChatGPT-4o)

Query: <How to calculate the carbon emissions of HPC in different segmentations>

Output:

The carbon emissions of hardware construction can be assessed by considering several factors including the materials used, the energy consumed during manufacturing, transportation, and end-of-life disposal. A common approach for calculating the carbon footprint of hardware production involves the following formula

Carbon Emissions Formula:

$$C_{\text{total}} = \sum C = C_{\text{mat}} + C_{\text{trans}} + C_{\text{usage}} + C_{\text{end}}$$

Where:

Material Emissions: Based on the types and amounts of materials used in the hardware (metals, rare earth elements, etc.), their extraction, modification, and manufacture.

Transport Emissions: Emissions generated from transporting raw materials to the manufacturing facility and the finished product to market.

Usage Emissions: Energy used while running the HPC system.

End-of-life Emissions: Emissions produced when the hardware reaches the end of its useful life (e.g., disposal, recycling, or landfill)

(2) Perplexity (Nov. 9, 2024 version, Perplexity Pro)

Query: <how much energy does an evaporative cooling system use per liter of water used?>

Output:

Based on the search results, I can calculate the energy efficiency of evaporative cooling systems:

Energy to Water Ratio

Evaporative cooling systems are highly efficient, with 1 cubic meter (1,000 liters) of water producing up to 695 kWh of cooling power[3]. This means:

$$\text{Energy Efficiency} = \frac{695 \text{ kWh}}{1000 \text{ L}} = 0.695 \text{ kWh/L}$$

Comparison with Traditional AC

Traditional air conditioning systems are significantly less efficient:

To produce the same 695 kWh of cooling capacity, traditional AC requires at least 250 kW of electrical input[3]

Modern evaporative coolers can achieve an Energy Efficiency Ratio (EER) up to 40, which is about 10 times higher than traditional cooling systems[3]

Power Consumption Examples

Small portable units (500 CFM) use less than 100 watts of electricity and about 4 liters of water per hour[5]

Larger systems can use between 41-115 liters per hour[1]

Advanced systems like IntrCool can deliver up to 40 kW of cooling power while consuming only 1 kW of electricity[3]

Citations:

[1] <https://www.rcdelectricalperth.com.au/blog/how-much-water-does-an-evaporative-cooler-use>

[2] https://climas.arizona.edu/sites/climas.arizona.edu/files/migrated_media/pdfwater-energy-trade-offs-between-swamp-coolers-and-air-conditioners_5.pdf

[3] <https://www.oxy-com.com/blog-news/evaporative-cooling-water-consumption>

[4] <https://www.aceee.org/files/proceedings/2012/data/papers/0193-000025.pdf>

[5] <https://alloutcool.com/swamp-cooler-water-energy-use/>

(3) Perplexity (Nov. 19, 2024 version, perplexity)

Query: <The proportion of each transportation (rail maritime road air) being used when transporting HPC (high performance computers)>

Output:

The proportion of each mode of transportation used for transporting High-Performance Computers (HPCs) can vary depending on factors such as the origin and destination, urgency, and cost considerations. While specific data may not be readily available for the transportation of HPCs alone, I can provide a general overview based on industry trends for shipping high-value, sensitive equipment.

General Proportions for Transportation of High-Value Electronics:

Air Transport (30-50%)

Proportion: Air is often used for high-value, time-sensitive shipments. HPCs, being high-cost and requiring careful handling, are frequently transported by air to ensure faster delivery and minimize risks of damage.

Reason: Quick delivery and lower risk of damage during transit. The value and urgency of the shipment justify the higher cost.

Road Transport (25-40%)

Proportion: Road transport is commonly used for shipments over short to medium distances or as part of the last-mile delivery.

Reason: Offers flexibility in routing and delivery, though it's slower compared to air. Road transport is also used in combination with other modes (e.g., air transport followed by road transport).

Rail Transport (10-25%)

Proportion: Rail transport might be utilized for longer distances, especially in regions with well-established rail networks, like Europe and parts of Asia.

Reason: More economical for bulk shipments over land, though slower than air or road. Rail is often chosen for non-urgent shipments that can withstand longer transit times.

Maritime Transport (5-20%)

Proportion: Maritime transport is the least likely mode for transporting HPCs due to the risk of damage, longer transit times, and handling requirements. However, it may be used when transporting large quantities or bulk shipments over long distances.

Reason: Cost-effective for large shipments but slower and requires additional handling on arrival at ports.

(4) OpenAI *ChatGPT* (Nov. 19, 2024 version, ChatGPT-4o)

Query: <The proportion of each transportation (rail maritime road air) being used when transporting HPC (high performance computers)>what is the proportion for HPC to choose for types? (Small, educational Medium, common technology advances Large, research institutions and colleges (extensive computational tasks) Supercomputers (large parallel processing tasks)>

Output:

When selecting High-Performance Computing (HPC) systems for different types of users or institutions, the proportion or capacity of the system typically aligns with the scale and complexity of their computational needs. Below is a general guideline for the type of HPC system each category would require:

Small (Educational)

Use case: Educational institutions with smaller computational needs, primarily for learning, basic research, and teaching.

System specs: Basic workstations or small clusters. Typically, fewer cores and lower memory capacity compared to larger systems.

Proportion: Limited in terms of processing power and scalability. Might use a modest number of nodes, typically with less parallelism.

Medium (Common Technology Advances)

Use case: Companies, small research teams, or tech startups looking to leverage HPC for moderate-scale simulations, modeling, and data analysis.

System specs: Mid-sized clusters with moderate processing power (e.g., several dozen to

a few hundred cores), often including GPUs for parallel processing.

Proportion: A more balanced system, offering reasonable scalability and performance for a variety of tasks (e.g., machine learning, data analysis, engineering simulations).

Large (Research Institutions and Colleges)

Use case: Larger research institutions or universities with more demanding computational tasks, such as complex simulations, large-scale data analysis, and scientific computations.

System specs: Larger clusters with a greater number of nodes and high scalability, likely including specialized hardware such as GPUs or accelerators. More memory, storage, and network bandwidth to handle big data and more complex simulations.

Proportion: High computational resources for research-grade tasks. These systems might include hundreds or thousands of cores and petabytes of storage.

Supercomputers (Large Parallel Processing Tasks)

Use case: Large-scale, high-performance computing tasks such as weather forecasting, climate modeling, large scientific research simulations, and artificial intelligence.

System specs: Supercomputers involve massive parallel processing, often with tens of thousands to millions of cores, high-speed interconnects, massive memory systems, and extensive storage. Typically, these are custom-built systems with state-of-the-art technology.

Proportion: These are the largest and most powerful systems, designed to handle the most computationally intensive and complex problems.

In terms of **proportion** in the overall HPC ecosystem:

Small systems would make up a very small fraction, usually in educational settings or small-scale research groups.

Medium systems are more common, as they suit a wide range of organizations, including startups and mid-sized research departments.

Large systems are still quite common but only in larger institutions or research centers with considerable computational demands.

Supercomputers represent the top end of the spectrum, typically a very small percentage of the total HPC systems globally, but they have immense impact in fields requiring massive parallel computation.

So, the distribution can roughly be described as:

Small: 10–20% (Educational or entry-level research)

Medium: 30–40% (Most common across industries)

Large: 20–30% (Research-heavy institutions or larger enterprises)

Supercomputers: 5–10% (Very high-performance systems for the most complex tasks)

These proportions can vary based on industry trends, the evolving computational needs of sectors like AI and data science, and investment in research and technology development.

(5) Perplexity (Nov. 9, 2024 version, Perplexity Pro)

Query: <The emission factors of transport of different transportations>

Output:

1. Road Transport

- Passenger Cars: Emission factors can range from 120 to 180 g CO₂e/km, depending on engine size and fuel type (petrol or diesel)
- Buses: Emissions are generally lower per passenger due to higher occupancy rates, averaging around 70 to 100 g CO₂e/km
- Trucks: Heavy-duty trucks can emit between 200 to 300 g CO₂e/km, influenced by load and driving conditions .

2. Rail Transport

- Freight Trains: Emission factors typically range from 20 to 40 g CO₂e/tonne-km, making rail one of the more efficient modes for freight transport
- Passenger Trains: These emit approximately 30 to 60 g CO₂e/passenger-km, depending on the train type and occupancy levels

3. Aviation

- Commercial Flights: Emission factors for air travel are significantly higher, averaging around 250 to 300 g CO₂e/passenger-km. This figure can increase when considering non-CO₂ effects such as contrails and radiative forcing, which can triple the climate impact of aviation emissions alone

4. Maritime Transport

- Cargo Ships: Emission factors for shipping vary widely but generally range from 10 to 30 g CO₂e/tonne-km, depending on vessel size and types.
- Ferries: These can have higher emissions due to their operational profiles, averaging around 50 to 100 g CO₂e/passenger-km

5. Non-Motorized Transport

- Bicycles: The emissions associated with cycling are minimal; estimates suggest around 10 g CO₂e/km when considering food energy consumption to sustain cyclists
- Walking: Emissions are negligible (approximately 0 g CO₂e/km) as it primarily involves human energy expenditure without significant carbon emissions